

From Concept to Submission in 8

An Agentic Research Workflow: Ridge-Cal, Digital Twins
and the Birth of a Statistical Research Skill

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The Tech Stack Journey

Setting Up the Infrastructure

Configured an AI agent ("Natasha") as my research assistant, connected via Telegram

Started with local models (LM Studio) - tool calling was unreliable

Switched to DeepSeek API: cheap (~\$0.15/M input), fast, reliable tool calling

Added Qwen via Ollama as a second model for independent cross-checking

Integrated Whisper so I could dictate ideas while pacing around the kitchen

Key Dependencies:

R + glmnet/furrr (simulations) | Python + matplotlib (diagrams)

Whisper (transcription) | Pandoc + LaTeX (PDF generation)

Lesson: start with the API that works, optimize later

The Spark: Where It Started

The Problem

PROCOVA is EMA-qualified for improving trial efficiency via prognostic scores

But it assumes the external score is perfectly calibrated for the trial population

Population shift is the rule, not the exception - especially in oncology

No existing method diagnoses or corrects miscalibration using blinded data

The Ridge-Cal Idea I Had:

Treat the external score as a pre-trained model, apply ridge-penalized Cox correction

Diagnose -> Calibrate -> Analyze. 6 parameters, blinded data, no new data collection

The 8-Hour Journey

Timeline: May 17-18, 2026

- ~21:00 Initial concept brainstorm: TMLE + Bayesian MCMC (too complex, scrapped)
- ~22:00 Pivot to ridge regression on 5 covariates
- ~23:00 First simulation - agent found a bug (non-PH assumption)
- ~00:00 10K-rep simulation running
- ~01:00 MAP-Cox bug surfaced (k applied after pooling)
- ~02:00 Bug fixed, re-simulation triggered
- ~06:00 Manuscript drafted - agent wrote prose, I reviewed every section
- ~07:00 Sent to Gemini (Reviewer 2) - Major Revision, 4 issues
- ~07:15 Agent fixed all 4 issues per my instructions
- ~08:00 Sent to Qwen for independent review - 7 MORE issues surfaced
- ~09:00 Digital twin landscape report drafted
- ~10:00 JBS formatting, self-verify, corrections
- ~14:00 Voice note while kitchen pacing - small strata investigation
- ~15:00 Workflow codified into reusable skill file

What Came Out of It

Deliverables in ~8 hours of active collaboration:

12-page Ridge-Cal manuscript - submission-ready for JBS

Response to Reviewer 2 - point-by-point rebuttal

Digital twin landscape survey (12 pages, 3 diagrams) - for SVPs

Small strata white paper with SAP language recommendations

Research Workflow skill file (SKILL.md) - reusable

All simulation code on GitHub (public repos)

The Multi-Model Review Loop

Why Not One AI?

Same-model reviewers share the same blind spots and hallucination patterns

Internal Loop (my OpenClaw agent, ~\$0.15/M tokens):

Primary work: DeepSeek v4 Flash | Independent reviewer: Qwen

Caught: Section 4 redundancy, delta justification, missing data, table headers

External Loop (separate AI, separate company):

Gemini Pro - completely different training, different blind spots

Caught: Non-collapsibility, tone, event rates, LoRA framing

The Convergence

Gemini: Major Revision -> Accept (1 round, 4 issues)

Qwen: Major -> Minor -> Accept (3 rounds, 7 DIFFERENT issues)

Total: 11 distinct bugs caught by 2 different AI models

Qwen caught things Gemini completely missed:

Section 4 redundancy, $\Delta = 0.01$ justification, table header ambiguity

MAP-Cox unfair framing, sandwich variance scope

Missing data acknowledgment, reference formatting

Takeaway: model-diverse review converges faster than any single reviewer

Demo: Voice Note to White Paper in 10

The Small Strata Investigation

Input: Me pacing the kitchen, talking to my phone:

"Do we need to pool small strata for CMH and MN methods?"

Whisper transcribed it on-device in < 1 min

- 1m** Problem scoping - I dictated, agent structured into Phase 0 proposal
- 1m** Independent Qwen review (isolated) - caught framing weakness
- 1m** I directed reframing as internal white paper
- 3m** Agent drafted R code, I reviewed specs, agent ran 5K reps
- 1m** Code review v1 - Qwen caught 3 bugs in our simulation
- 1m** Code review v2 - agent verified fixes, I confirmed
- 1m** Final review - white paper compiled, ready
- <1m** Pushed to public GitHub repo

Small Strata Results

Method	Failure Rate	Type I	
CMH OR (+0.5 CC)	0.000	0.043-0.054	
CMH RR (GR var)	0.000	0.045-0.081	
Stratified MN RD	0.000	0.035-0.069	
Cox PH (stratified)	N/A	N/A	
Log-rank (stratified)	N/A	N/A	

Conclusion: No pooling required for any method.

Code reviews caught: Scenario 4 duplicated, RR variance unstratified, Wald mislabeled as MN.

All bugs fixed before delivery - independent AI review caught what we overlooked.

The Research Workflow That Eme

Codified into a Reusable Skill File - 8 Phases

Ph 0: Topic ID	Identify gap, map contradictory literature
Ph 1: Writeup	Write, spawn isolated reviewer, iterate
Ph 2: Simulate	2 -> 20 -> 200 -> 10,000 reps (progressive)
Ph 3: QC	CODE REVIEW BEFORE BIG RUNS - mandatory checkpoint
Ph 4: Full Run	Background, scheduled, monitored
Ph 4.5: Audit	Verify code matches manuscript claims
Ph 5: Manuscript	Write, verify refs, PDF self-check, senior review
Ph 6: Revision	Parse -> Reconcile -> Re-sim -> Diff check
Ph 7: Submit	Independent reviewer -> Revise -> Max 3 rounds

Key Lessons Codified

1. Isolate reviewers - fresh sessions, no cross-contamination
2. Diversify models - different AIs catch different bugs
3. Code review BEFORE big simulations - trust but verify
4. PDF self-verify - programmatic checks before sending to humans
5. Clean up cruft every iteration - stale files cause confusion
6. Push to GitHub at milestones - enables external review
7. Batch deliveries - dont send incremental fixes, pace yourself

Key Takeaways

What Worked Well:

Independent AI reviewers caught 11 bugs before human eyes saw the draft

Small tests before big runs - avoided ~5 dead ends

Human-in-the-loop essential - I directed 8 reframing cycles

The agent amplified judgment, did not replace it

What We Would Do Differently:

PDF generation was painful - 5 failed attempts, silent image failures

Now has a verification script baked into the pipeline

Code review almost skipped - hard checkpoint now

Stale artifacts accumulated - explicit cleanup at milestones

The Bottom Line

Session cost: ~500K DeepSeek tokens, well under \$1.00

That bought:

- 1 manuscript ready for JBS submission
- 1 SVP-ready digital twin report (12 pages, 3 diagrams)
- 1 complete simulation study + white paper + SAP language
- 1 reusable research workflow skill
- 3 independent AI reviews from 2 different model architectures
- 3 GitHub repos with full code

Key insight:

The agent doesn't replace the statistician - it replaces the drafting lag and review cycle

What used to take a week of calendar coordination now takes an evening

Thank You

Questions?

Repos: github.com/doublerobust/ridge-cal | github.com/doublerobust/small-strata-p

The skill: SKILL.md in my OpenClaw workspace - available on request